

Dr. Mujeeb Ahmad
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Academic Profile

Experimental condensed matter physicist with expertise in thermoelectric materials, topological quantum materials, low-temperature quantum transport, and nanoscale thin-film devices. Experienced in advanced thin-film growth, nanofabrication, thermoelectric transport, and quantum thermal transport measurements. Research interests focus on topological thermoelectric materials, 2D–3D heterostructures, and energy-efficient quantum devices for future nanoelectronic and energy harvesting technologies.

Research Interests

Topological Quantum Materials • Thermoelectric Materials • Quantum Transport • Weyl and Dirac Semimetals • Thin-Film Nanotechnology • 2D–3D Heterostructures • Nanoscale Heat Transport • Spin–Orbit Coupling • Energy Harvesting Devices • Quantum Thermal Transport • Nanofabrication and Device Physics

Professional Profile

- More than 6 years of research experience in thermoelectric materials, topological quantum materials, and low-temperature quantum transport phenomena.
- Extensive expertise in electrical, thermal, and thermoelectric transport measurements in Weyl semimetals, Dirac materials, and low-dimensional systems.
- Strong hands-on experience in thin-film growth techniques including RF/DC sputtering, Pulsed Laser Deposition (PLD), chemical vapor deposition (CVD), and hydrothermal synthesis.
- Expertise in advanced nanoscale characterization techniques including AFM, KPFM, MFM, scanning thermal microscopy (SThM), conductive AFM (CAFM), XRD, Raman spectroscopy, and PPMS-based transport measurements.
- Hands-on experience in cleanroom nanofabrication including photolithography, maskless lithography, e-beam lithography, spin coating, thermal evaporation, and rapid thermal annealing.
- Developed and designed a variable-temperature thermal conductivity measurement system based on the 3ω method for thin-film structures.
- More than 20 peer-reviewed publications in leading international journals including Advanced Functional Materials, Physical Review B, ACS Applied Materials & Interfaces, Applied Physics Letters, and Journal of Applied Physics.
- Active reviewer for international journals including Advanced Functional Materials, ACS Applied Materials & Interfaces, Physical Review B, and Sensors and Actuators B: Chemical.
- Strong skills in **scientific communication, research leadership, experimental design, data analysis, and project organization.**

Professional Experience

Institute of Physics, Polish Academy of Sciences Warsaw, Poland

Aug 2021- Present

Postdoctoral Fellow

Mentor: [Prof. Maciej Zgirski](#), Institute of Physics

Group: [International Centre for Interfacing Magnetism and Superconductivity with Topological Matter](#)

- Investigating quantum transport and thermal transport phenomena in topological quantum materials and superconducting systems.
- Performing low-temperature measurements on superconducting Josephson junctions and topological materials using dilution refrigerator systems.
- Studying electron-phonon interactions, vortex transport, quasiparticle dynamics, and topological transport phenomena.
- Conducting magnetotransport, thermoelectric, and thermal conductivity measurements in Weyl and Dirac semimetals.
- Maintaining and optimizing cryogenic and microwave measurement systems for quantum device characterization.
- Mentoring Ph.D. students in advanced low-temperature experimental techniques and transport measurements.

Indian Institute of Technology, Delhi, India

Oct 2020- July 2021

Senior Project Scientist

Mentor: [Prof. Bodh Raj Mehta](#), Department of Physics

Project: [Synthesis of Size-Selected Metal Nanoparticles by Physical and Chemical Methods for Plasmonic Applications](#)

- Performed FDTD simulations to demonstrate metal-coated SiO_2 nanospheres (30-100 nm diameter) with required plasmonic properties in the 900-1064 nm wavelength range, optimising parameters including metal nanostructure size, shape, thickness and nanosphere dimensions.
- The simulated structures were synthesised experimentally using chemical synthesis and physical vapour deposition techniques.
- The growth of bilayers/composites of metal nanostructures (on SiO_2 nanospheres) and the investigation of their plasmonic properties.

Education

Ph.D (2015-2020)

Department of Physics, Indian Institute of Technology, Delhi, India.

Mentor: [Prof. Bodh Raj Mehta](#).

Thesis Title: Synthesis, structural and thermoelectric properties of $MoS_2 : Sb_2Te_3$ and other 2D-3D nanocomposite and multilayer structures.

- Conducted research on 2D-3D nanocomposite and multilayer thin film structure for enhancing thermoelectric performance.
- Developed a method to study the nature of 2D-3D interfaces using scanning probe microscopy techniques and investigate their thermoelectric and electrical transport properties.
- Designed and developed thermal conductivity measurement system under high vacuum with temperature range 300°K to 500°K based on 3ω method for thin film structure.
- Fabricated a thermoelectric device using fabrication tools and observed 80 mV open circuit voltage at 45K temperature difference.
- Mentored three M. Tech, two M. Sc. students for their Master's thesis.

M.Sc. (Physics) (2011-2013)

Department of Physics, Indian Institute of Technology, Hyderabad, India.

CGPA:8.66

Mentor: [Prof. Prem Pal](#).

Thesis Title: A study of the precise alignment of mask patterns to the crystallographic orientation of silicon wafers.

B.Sc. (Physics) (2008-2011)
 Department of Physics, Aligarh Muslim University, Aligarh India.

Selected Research Contributions

- Demonstrated enhanced thermoelectric efficiency in 2D–3D multilayer heterostructures with a high figure of merit ($ZT \approx 2.02$ at room temperature).
- Investigated topological thermoelectric transport in ultrathin Bi_2Te_3 thin films with enhanced Seebeck coefficients due to spin–orbit coupling and topological surface states.
- Studied anomalous Hall and anomalous Nernst effects in magnetic Weyl semimetals driven by Berry-curvature transport phenomena.
- Investigated gravitational anomaly and magnetothermal transport in ferrimagnetic Weyl semimetals.
- Explored anisotropic phononic thermal transport and spin–phonon coupling in topological Weyl systems.
- Developed thermal conductivity measurement systems and nanoscale characterization methodologies for thin-film thermoelectric materials.

Selected Publications

- **Mujeeb Ahmad**, Md Shahin Alam, Xiaohan Yao, Fazel Tafti, Marcin Matusiak, “Anisotropic suppression of the phononic thermal conductivity by magnetic field in SmAlSi” *Phys. Rev. B.* 111, L161121, 2025.
- Md Shahin Alam, Alexandr Kazakov, **Mujeeb Ahmad**, Rajibul Islam, Fei Xue, Marcin Matusiak, “Quantum transport properties of the topological Dirac semimetal α -Sn” *Phys. Rev. B.* 109, 245135, 2024.
- Abhishek Ghosh, Chandan Kumar Vishwakarma, Prashant Bisht, Narinder Kaur, **Mujeeb Ahmad**, Bodh Raj Mehta, “Understanding the Role of an Oxygen-Passivated Grain Boundary in Modulating the Thermoelectric Properties of a Sb_2Te_3 Thin Film” *ACS Applied Energy Materials* 7, 5107-5119, 2024.
- Pardeep Kumar Tanwar, **Mujeeb Ahmad**, Md Shahin Alam, Xiaohan Yao, Fazel Tafti and Marcin Matusiak, “Gravitational anomaly in antiferromagnetic topological Weyl semimetal NdAlSi” *Phys. Rev. B.* 108, L161106, 2023.
- Md Shahin Alam, Amar Fakhredine, **Mujeeb Ahmad**, PK Tanwar, Hung-Yu Yang, Fazel Tafti, Giuseppe Cuono, Rajibul Islam, Bahadur Singh, Artem Lynnyk, Carmine Autieri and Marcin Matusiak, “Sign change of anomalous Hall effect and anomalous Nernst effect in the Weyl semimetal CeAlSi” *Phys. Rev. B.*, 107, L085102, 2023.
- **Mujeeb Ahmad**, Khushboo Agarwal, Sergio Gonzalez Munoz, Abhishek Ghosh, Nisha Kodan, Oleg Victor Kolosov, and Bodh Raj Mehta, “Engineering Interfacial Effects in Electron and Phonon Transport of Sb_2Te_3/MoS_2 Multilayer for Thermoelectric ZT Above 2.0” *Adv. Funct. Mater.*, 2206384, 1-13, 2022.
- Pardeep Kumar Tanwar, Md Shahin Alam, **Mujeeb Ahmad**, Marcin Kaczorowski, Dariusz and Matusiak “Severe violation of the Wiedemann-Franz law in quantum oscillations of NbP” *Phys. Rev. B.* 106, L041106, 2022.
- Prashant Bisht, Arvind Kumar, Ingvald T Jensen, **Mujeeb Ahmad**, Branson D Belle, Bodh Raj Mehta “Enhanced gas sensing response for 2D $\alpha - MoO_3$ layers: thickness dependent changes in defect concentration, surface oxygen adsorption, and metal-metal oxide contact” *Sensors & Actuators B: Chemical*, 341, 129953, 2021.
- Vishakha Kaushik, **Mujeeb Ahmad**, Pintu Das, Bodh R. Mehta, “Understanding the Role of 2D Nature on the Junction Properties in WS_2 Layers; Effect of AFM Tip Induced Loading Force on Spatially Varying Contact” *Surface and Interfaces*, 24, 101131, 2021.
- Abhishek Ghosh, **Mujeeb Ahmad**, Prasant Bisht and B. R. Mehta, ” Modifying the thermoelectric transport of Sb_2Te_3 thin film by incorporating size-selected gold nanoparticles having narrow size distribution and controlled concentration via carrier filtering effect ” *ACS Applied Material & interface*, 13, 13226-13234, 2021.
- Vishakha Kaushik, **Mujeeb Ahmad**, Khushboo Agarwal, Deepak Varandani, Branson D. Belle, Pintu Das, Bodh R. Mehta “Charge Transport in 2D MoS_2 , WS_2 and $MoS_2 - WS_2$ Heterojunction Based Field Effect Transistors; Role of Ambipolarity” *The Journal of Physical Chemistry C*, 124, 23368, 2020

- N. Kodan, **Mujeeb Ahmad** and B. R. Mehta, “Enhanced photoelectrochemical response and stability of hydrogenated ZnO nanorods decorated with Fe_2O_3 nanoparticles,” *Journal of Alloys and Compounds*, 34, 035026, 2019.
- N. Kodan, **Mujeeb Ahmad** and B. R. Mehta, “Charge carrier separation and enhanced PEC properties of $BiVO_4$ based heterojunctions having ultrathin overlayers,” *International Journal of Hydrogen Energy*, 46, 189-196, 2021.
- **Mujeeb Ahmad**, N. Kodan, A. Ghosh and B. R. Mehta, “The nature of 2D:3D SnS: Bi_2Te_3 interface and its effect on enhanced electrical and thermoelectric properties,” *Journal of Alloys and Compounds*, 847, 156233, 2020.
- **Mujeeb Ahmad**, K. Agarwal and B. R. Mehta, “An anomalously high Seebeck coefficient and power factor in ultrathin Bi_2Te_3 film; Spin-orbit interaction,” *J. Appl. Phys.*, 128, 035108, 2020.
- **Mujeeb Ahmad**, D. Varandani and B. R. Mehta, “Large surface charge accumulation in 2D MoS_2/Sb_2Te_3 junction and its effect on junction properties: KPFM based study,” *Appl. Phys. Lett.*, 113, 141603, 2018.

Academic Metrics

- Total Publications: 20+
- First-Author Publications: 5+
- Conference Presentations: 10+
- Publications in High-Impact Journals: Advanced Functional Materials, Physical Review B, ACS Applied Materials & Interfaces, Applied Physics Letters.
- Reviewer for: Advanced Functional Materials, Physical Review B, ACS Applied Materials & Interfaces
- Research Areas: Thermoelectric Materials, Topological Quantum Materials, Quantum Transport, Thin-Film Nanotechnology
- h-index: 14
- Citations: 486+

Awards, Prize and Grants

- May 2019, **Research Excellence Travel Grant** (150,000 INR)
Funding Agency: Indian Institute of Technology Delhi, India.
- Oct 2017, **Best Poster Award** International Workshop on Physics of Semiconductor Devices (IWPSD)
Indian Institute of Technology Delhi, India.
- Mar 2017, **Best Poster Award** Research Scholar Day
Department of Physics Indian Institute of Technology Delhi, India.
- June 2017, **International Travel Grant** (150,000 INR)
Funding Agency: Indian Institute of Technology Delhi, India.
- May 2016, **Best Poster Award** Research Scholar Day
Department of Physics Indian Institute of Technology Delhi, India.
- Jan 2015-Dec 2020 **Graduate Aptitude Test in Engineering (GATE) fellowship**
Funding Agency: Ministry of Human Resource Development (MHRD), India.

Technical Skills

- **Material Characterization Tools:** X-ray Diffraction System (Panalytical Xpert Pro), Scanning Probe microscopy (Bruker ICON system), LSR-3 Seebeck and electrical conductivity measurement system, Electrical and Magnetic measurement system (PPMS), Thermal conductivity measurement system (based on 3ω method), Electrical and thermal measurements performed in Cryostate system
- **Fabrication Tools:** High vacuum RF/DC sputtering systems, Thermal evaporation system, Chemical Vapour deposition system, E-beam lithography, Maskless photolithography, Pulsed Laser Deposition (PLD)
- **Other Expertise:** Clean room experience, Liquid nitrogen and helium handling experience

- **Data Analysis:** XPS, Raman Spectroscopy
- **Software and Programming:** MATLAB, LabVIEW, LaTeX

Research Facilities and Experimental Expertise

Experienced in the synthesis, fabrication, and characterization of low-dimensional and topological quantum materials using advanced thin-film growth and nanofabrication techniques. Hands-on expertise in RF/DC sputtering, Pulsed Laser Deposition (PLD), chemical vapor deposition (CVD), hydrothermal synthesis, thermal evaporation, and cleanroom-based fabrication processes including maskless photolithography, e-beam lithography, and rapid thermal annealing.

Extensive experience in low-temperature quantum transport, thermoelectric, and thermal conductivity measurements using PPMS systems, cryostats, dilution refrigerator systems, and customized transport measurement setups. Developed and designed a variable-temperature thermal conductivity measurement system for thin-film structures based on the 3ω method under high-vacuum conditions.

Strong expertise in advanced nanoscale and structural characterization techniques including atomic force microscopy (AFM), Kelvin probe force microscopy (KPFM), magnetic force microscopy (MFM), conductive AFM (CAFM), scanning thermal microscopy (SThM), X-ray diffraction (XRD), Raman spectroscopy, and XPS analysis. Experienced in studying charge transport, spin-orbit coupling, thermoelectric transport, and topological transport phenomena in Weyl semimetals, Dirac materials, and multilayer thin-film heterostructures.

Teaching Interests

Undergraduate Courses

- Solid State Physics
- Quantum Mechanics
- Introduction to Nanoscience and Nanotechnology

Postgraduate Courses

- Advanced Condensed Matter Physics
- Thermoelectric Materials and Devices
- Quantum Materials and Topological Physics
- Thin-Film Deposition and Nanofabrication
- Electronic Transport in Low-Dimensional Systems
- Materials Characterization Techniques

Languages

- English, Hindi, Urdu